Serge Guy Nana Engo
Department of Physics, Faculty of Science
University of Yaoundé I
P.O. Box 812, Yaoundé, Cameroon
serge.nana-engo@facsciences-uy1.cm

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The Editors
Communications in Nonlinear Science and Numerical Simulation

Dear Editors,

We submit the manuscript entitled "Synthetic-flux control of hyperchaos order in a dissipative optomechanical model" for consideration as a full-length research article in *Communications in Nonlinear Science and Numerical Simulation*.

The central question we address is whether a gauge-invariant synthetic-flux phase Φ_{syn} can act as a reproducible open-loop control coordinate for the *order* of a drive- and coupling-gated hyperchaos transition in a dissipative optomechanical model. Within a normalized six-dimensional model (one cavity plus two mechanical resonators), the answer is yes at the model level: the full six-exponent Lyapunov spectrum resolves up to four simultaneously positive Lyapunov directions, beyond the order of two reported for quadratic single-mode optomechanics by Halef and Shomroni, *Phys. Rev. Lett.* **135**, 257201 (2025). The integer count $n_+ = 4$ is attained only at selected flux phases (near the quadratures) and remains discretization- and threshold-sensitive at marginal points. The route proceeds via a supercritical Neimark–Sacker bifurcation at $E^* = 1.060$ ($\theta = 0$) and $E^* = 0.975$ ($\theta = \pi/2$), with re-stabilization at the sampled intermediate drive $E = 2.0$ and finite-time positive-exponent counts reaching three at $E = 4.0$ and four at $E = 8.0$.

Our study contributes to the nonlinear-science methodology in three ways. First, it establishes a *phase-consistent Lyapunov–Floquet protocol* for classifying the order of a hyperchaos transition, combining monodromy multipliers, full QR/Benettin spectra, and the analytic dissipative trace into a single consistency criterion. Second, it localizes the onset of the hyperchaos route to a Neimark–Sacker torus bifurcation via drive-axis continuation of the unstable periodic orbit, providing a concrete bifurcation-route picture rather than a grid-level classification alone. Third, it provides a *matched-measurement Fisher bound* showing that additional unstable directions do not automatically yield metrological gain under shared resources—a geometric statement that complements the extensive CNSNS literature on hyperchaos control and synchronization [M. Rafikov and J. M. Balthazar, *Commun. Nonlinear Sci. Numer. Simul.* **13**, 1246–1255 (2008); doi:10.1016/j.cnsns.2006.12.011].

Confidence in the classification rests on a phase-consistent Floquet–Lyapunov protocol: monodromy multipliers, dissipative volume balance $\sum_i \lambda_i = \mathrm{Tr}(J) = -1.04$, a 432-run classification ensemble spanning twelve seeds, twelve flux phases, and three drives, and orbit continuation with window-convergence tests. As a secondary geometric clarification, the matched-resource force-sensing protocol yields only a modest enhancement on the stable reference orbit (1.039 ± 0.014 flux-off, 1.019 ± 0.007 single-mode) and a null result on the chaotic attractor (flux-on Fisher information $0.54\times$ flux-off at the resolved $E = 4$ points); this null is specific to the matched-resource protocol and is presented as a geometric bound, not as a general verdict on chaotic sensing.

All results are strictly model-level: the explored sector lies $2542\times$ beyond anchored silicon couplings, and the robust hyperchaos onset sits near $g_1/\omega_m \approx 0.18$ ($1525\times$ the anchor). Closing the gap requires ultrasonic characterization of the mechanical degeneracy ω_2, a measured inter-resonator hopping J_m, and a genuine gigahertz platform. Four limitations bound the claims: (i) the classification ensemble does not establish uniqueness, and $n_+ = 4$ remains threshold- and seed-sensitive at marginal phases; (ii) the null sensing result is specific to the matched-resource constraint; (iii) Fock- and Wigner-level checks do not bound quantum corrections to the Lyapunov spectrum or to the Fisher information; (iv) the strong-coupling sector lies far above present experimental anchors. We make no claim of quantum advantage, standard-quantum-limit beating, or experimental validation.

The manuscript fits the journal's scope on bifurcations, attractors and chaos, high-dimensional chaos, and Lyapunov analysis and control. The control studied is open-loop and parametric; the matched-measurement analysis is a geometric bound rather than a device claim. The manuscript is accompanied by a Supplemental Material containing the full numerical convergence records, the formal proofs of the consistency criterion and matched-Fisher lemma, and the per-parameter feasibility tables. Manuscript sources, deterministic analysis scripts, and the environment specification are publicly archived at https://github.com/NanaEngo/floquet-chaos-synthetic-flux.

This work is original, has not been published previously, and is not under consideration elsewhere. All authors have approved the manuscript and declare no competing interests. No funding was received. Thank you for your consideration.

Sincerely,

Serge Guy Nana Engo
on behalf of all authors